

An Integrated Security Service System for 5G Networks using an I2NSF Framework

[draft-ahn-nmrg-5g-security-i2nsf-framework](#)

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IETF-126 5G-I2NSF System for Integrated Security Services in 5G Networks

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IETF-126 Vienna
18-24 July 2026

IETF-126 5G-I2NSF Project

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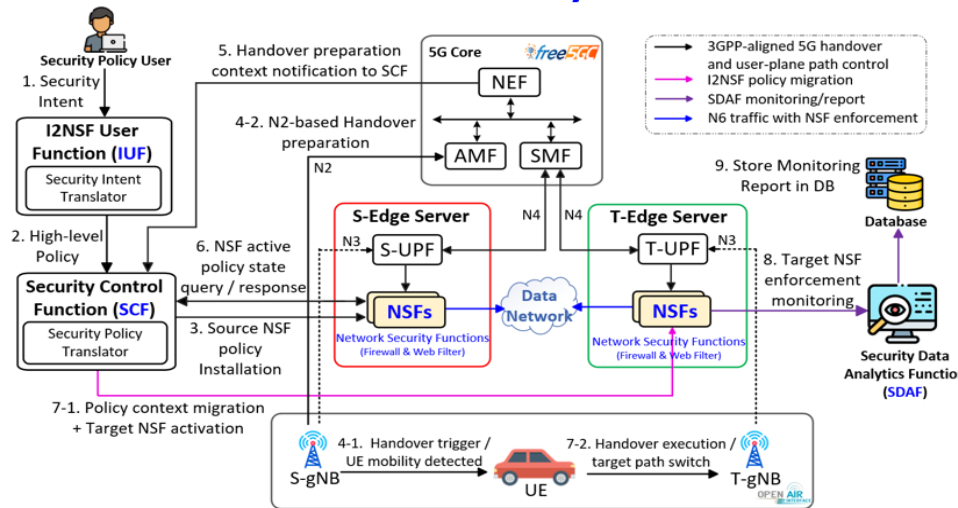
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5G-I2NSF System



Objectives

- This project develops a 5G Core-assisted I2NSF framework that coordinates security policy orchestration with the 3GPP-aligned N2-based handover procedure.
- The proposed system proactively migrates the verified security policy context from the Source NSF to the Target NSF before UE traffic is switched from the S-UPF path to the T-UPF path.
- Edge-deployed NSFs, co-located with UPFs, enforce the migrated policy on the N6 traffic path, while SDAF monitors the enforcement result for closed-loop validation.

Future Work

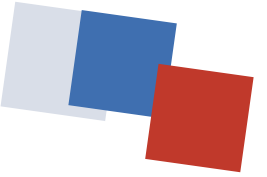
- Integrate AI-based intent interpretation with SDAF feedback analysis for automated policy refinement and security service management.

5G-I2NSF Development Environment

- OS: Ubuntu 24.04
- Free5GC VM: version 4.2.3
- OpenAirInterface 5G (OAI): 2026.w26
- GitHub Repository:
<https://github.com/jaehoonpauljeong/5G-I2NSF/tree/main/IETF-126>

Workflow of the 5G-I2NSF Testbed

1. The Security Policy User submits a security intent to the I2NSF User Function (IUF).
2. The IUF translates the security intent into a high-level policy and forwards it to the Security Control Function (SCF).
3. The SCF translates the high-level policy into enforceable NSF policies and installs the Source NSFs at the S-Edge Server.
4. When UE handover is triggered, the 5G Core performs N2-based handover preparation, and the SMF prepares the UPF path via N4.
5. The 5G Core exposes the handover preparation context, including UE mobility and target path information, to the SCF through the NEF-compatible interface.
6. The SCF checks the currently running NSFs and retrieves the active policy based on their operational state before policy migration.
7. The SCF migrates the active policy from the S-Edge Server to the T-Edge Server and activates the Target NSFs before UE traffic is switched.
8. The SDAF monitors whether the Target NSFs correctly enforce the migrated policy on the N6 traffic path or not.
9. The SDAF stores the Target NSF monitoring results in the database for later validation, analysis, and policy refinement.



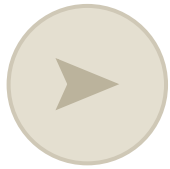
Motivation and Goal

➤ Motivation

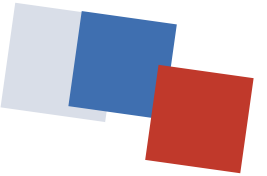
- 5G edge networks require security services to be deployed close to user-plane traffic paths to support ultra-low-latency communication.
- A distributed and intent-based security framework is needed to automate policy translation and enforcement without introducing significant security service delay.

➤ Goal

- Develop and demonstrate an edge-based 5G security management system based on the IETF I2NSF framework.
- Based on the active NSF and its policy context in the Source Edge Server, determine and activate the corresponding NSF in the Target Edge Server before the user-plane path switch.



An Integrated Security Service System for 5G Networks using an I2NSF Framework (1/2)



- An Integrated Security Service System for 5G Networks using an I2NSF Framework
 - <https://datatracker.ietf.org/doc/draft-ahn-nmrg-5g-security-i2nsf-framework/>

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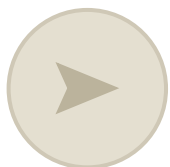
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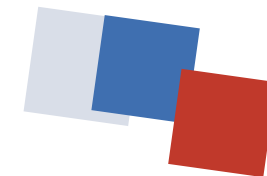
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An Integrated Security Service System for 5G Networks using an I2NSF Framework

Abstract

This document presents an integrated framework for automated security management in 5G edge networks using the Interface to Network Security Functions (I2NSF) architecture. The proposed system leverages Intent-Based Networking (IBN) to allow users or administrators to declare high-level security intents, which are translated into enforceable network and application policies. Network-level policies are delivered to 5G core components via the Network Exposure Function (NEF), while application-level policies are enforced directly on a User Equipment (UE) through distributed IBN Controllers. This architecture supports adaptive, context-aware, and distributed policy enforcement, enabling real-time response to dynamic edge conditions and user mobility scenarios such as handovers. By integrating closed-loop monitoring and analytics, the system ensures consistent and autonomous security across heterogeneous 5G environments.

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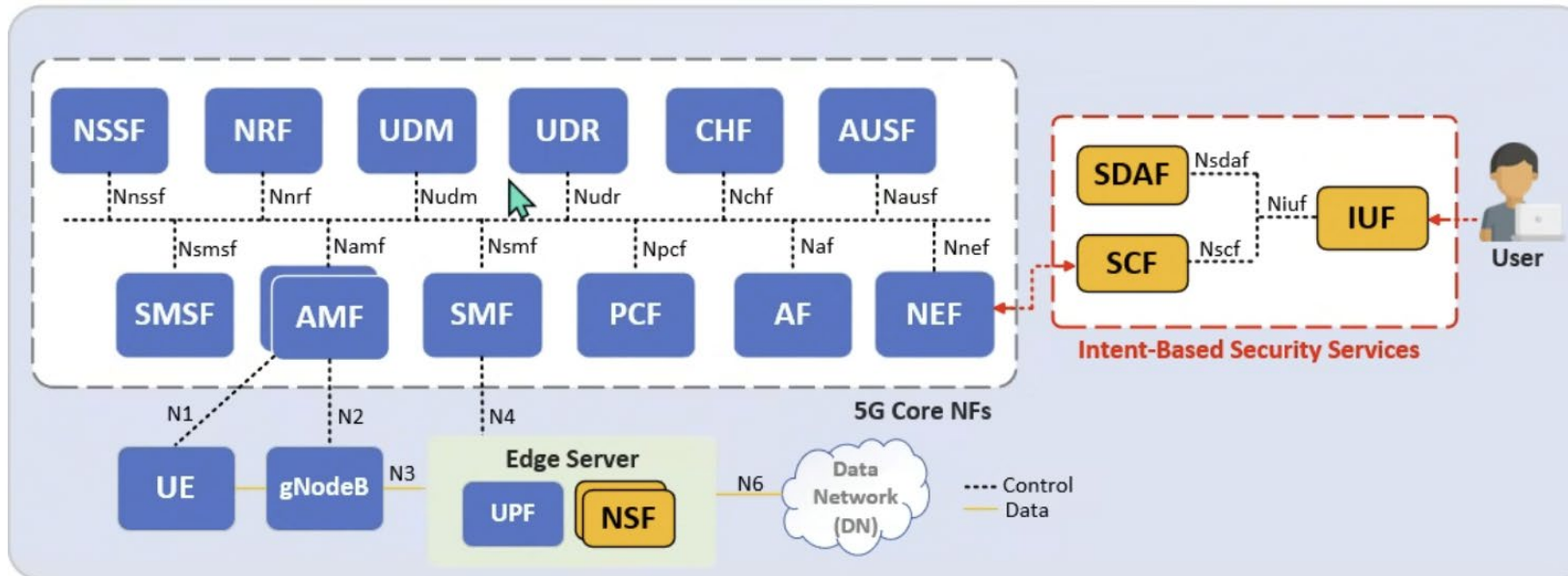
1. Introduction

Network softwarization has become a fundamental approach for delivering network services across various infrastructures, including 5G mobile networks [TS-23.501], cloud computing platforms, and edge computing environments. This paradigm is enabled through key technologies such as Network Functions Virtualization (NFV) [ETSI-NFV] and Software-Defined Networking (SDN) [RFC7149]. In addition, Intent-Based Networking (IBN) [RFC9315] [Survey-IBN-CST-2023] serves as a foundation for implementing intelligent behaviors in both network-level and application-level services. As networks continue to evolve in this software-driven direction, the emergence of 5G introduces new challenges, particularly in the realm of security.

As mobile networks evolve toward 5G, the increasing complexity of network functions and the widespread deployment of edge devices such as IoT nodes, user equipment (UE), and application functions (AFs) [TS-23.501] introduce significant challenges to existing security models. These environments are inherently dynamic, heterogeneous, and latency-sensitive, making it difficult for traditional rule-based configurations, which are typically static and manually managed, to respond effectively to changing conditions. In particular, security

Security Management Framework for 5G Edge Networks

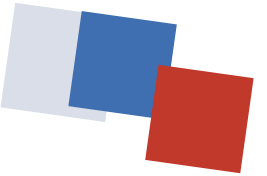
➤ Intent-driven security framework for automated and adaptive 5G edge security



- **Intent-Based Security Services**
Converts intents into security policies.
- **5G Core Integration**
Delivers translated policies to 5G Core functions through NEF.
- **Edge Security Enforcement**
Enforces security policies near UPFs using distributed NSFs.
- **Closed-Loop Security Management**
Monitors policy enforcement and refines security decisions.

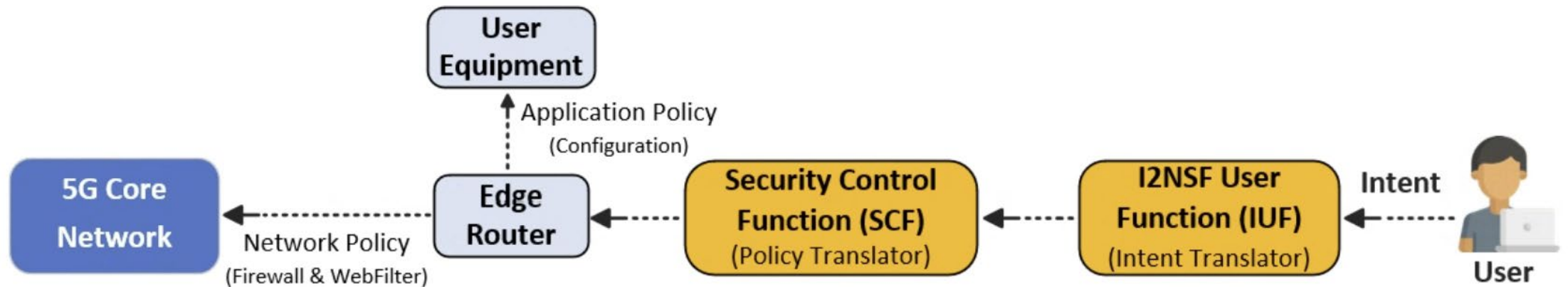


Intent Translation and Policy Delivery



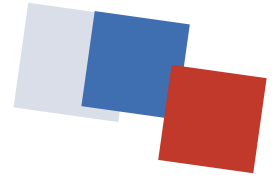
➤ Intent-based Integrated Security Management

- User intent is translated into network and application security policies.
- The Security Control Function coordinates policy enforcement between the 5G Core Network and the User Equipment.
- This enables unified security control across the edge network and end devices.



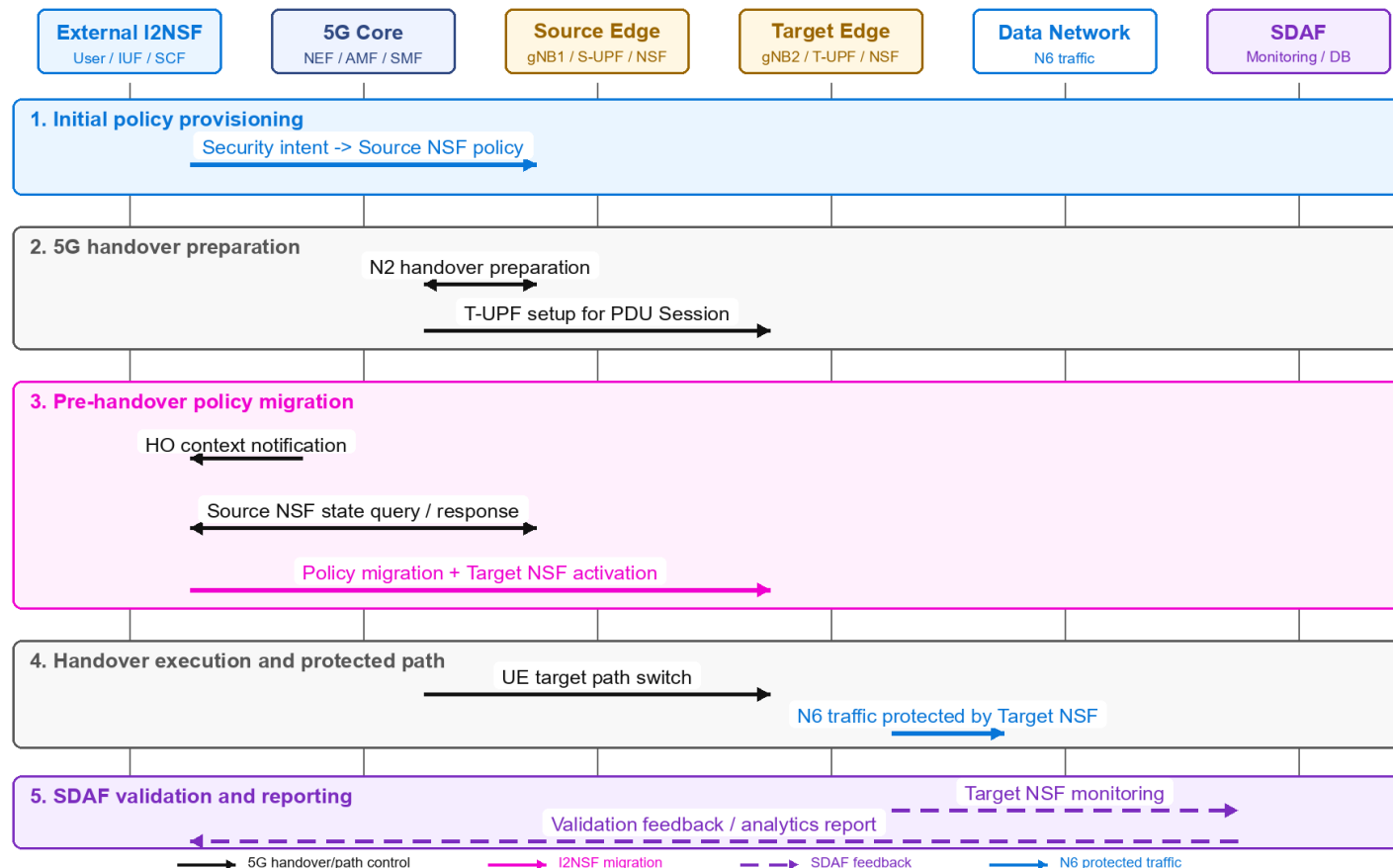


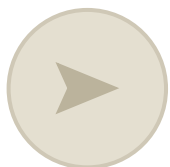
I2NSF-based Integration for 5G Security Services



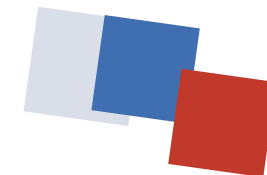
➤ Mobility-Aware Security Policy Migration during 5G Handover

- Proactive security policy migration enables the Target NSF before the UE path switch, ensuring continuous traffic protection throughout the 5G handover.

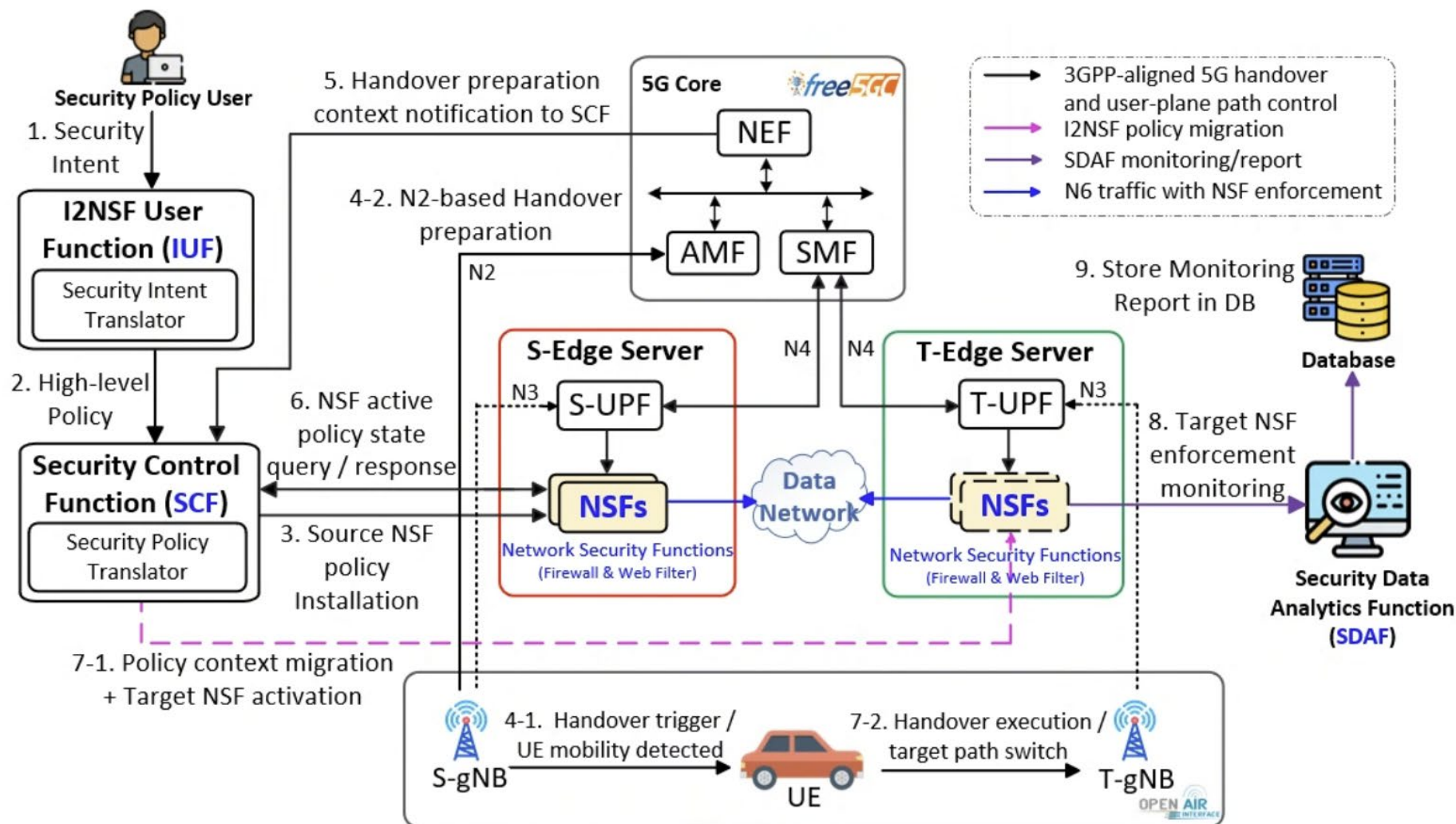




An Integrated Security System for 5G Networks

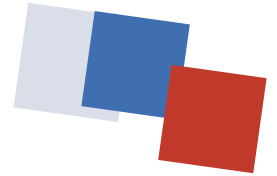


➤ 5G-I2NSF System Architecture

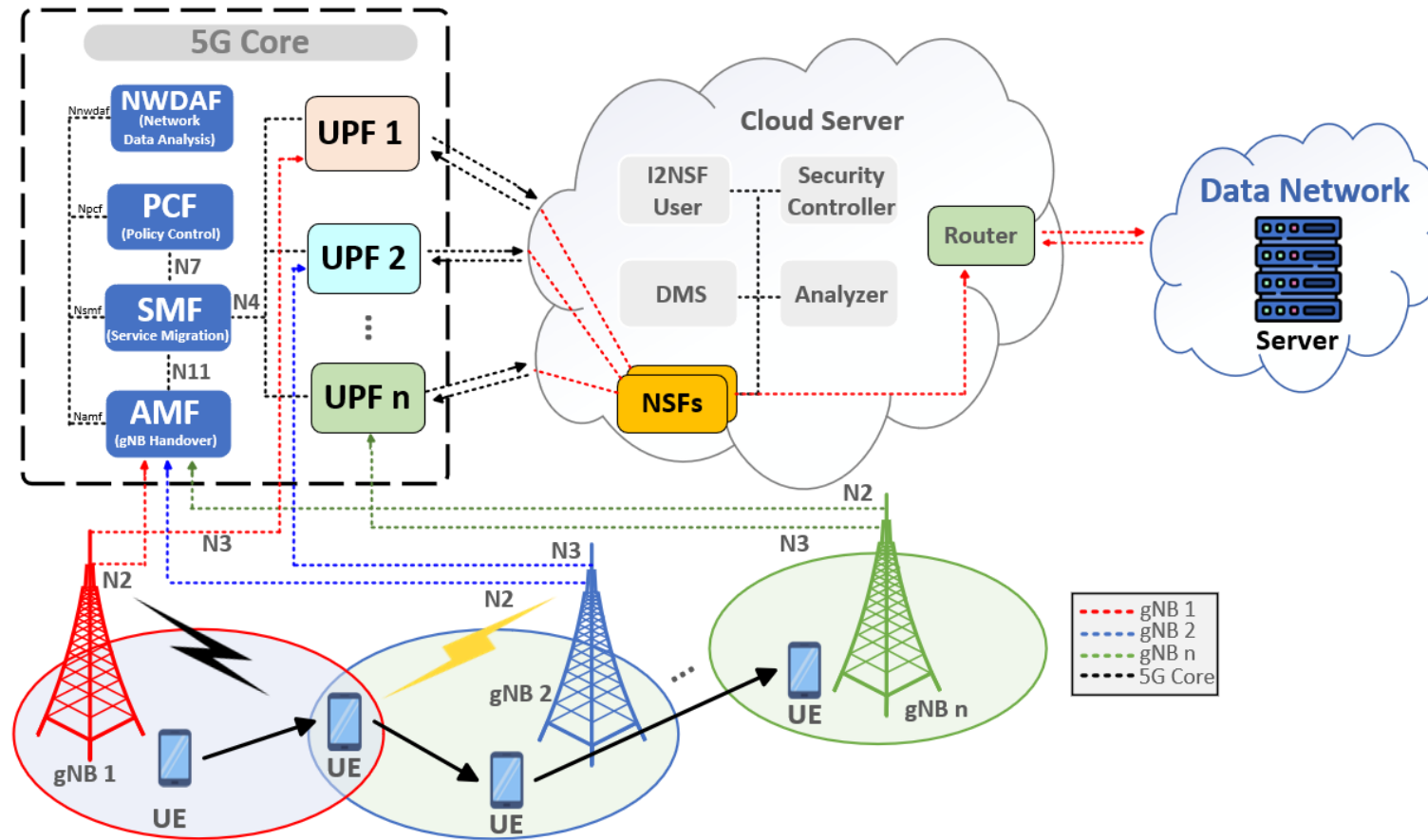


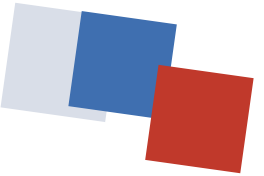


Cloud-based Security Approach for 5G Networks



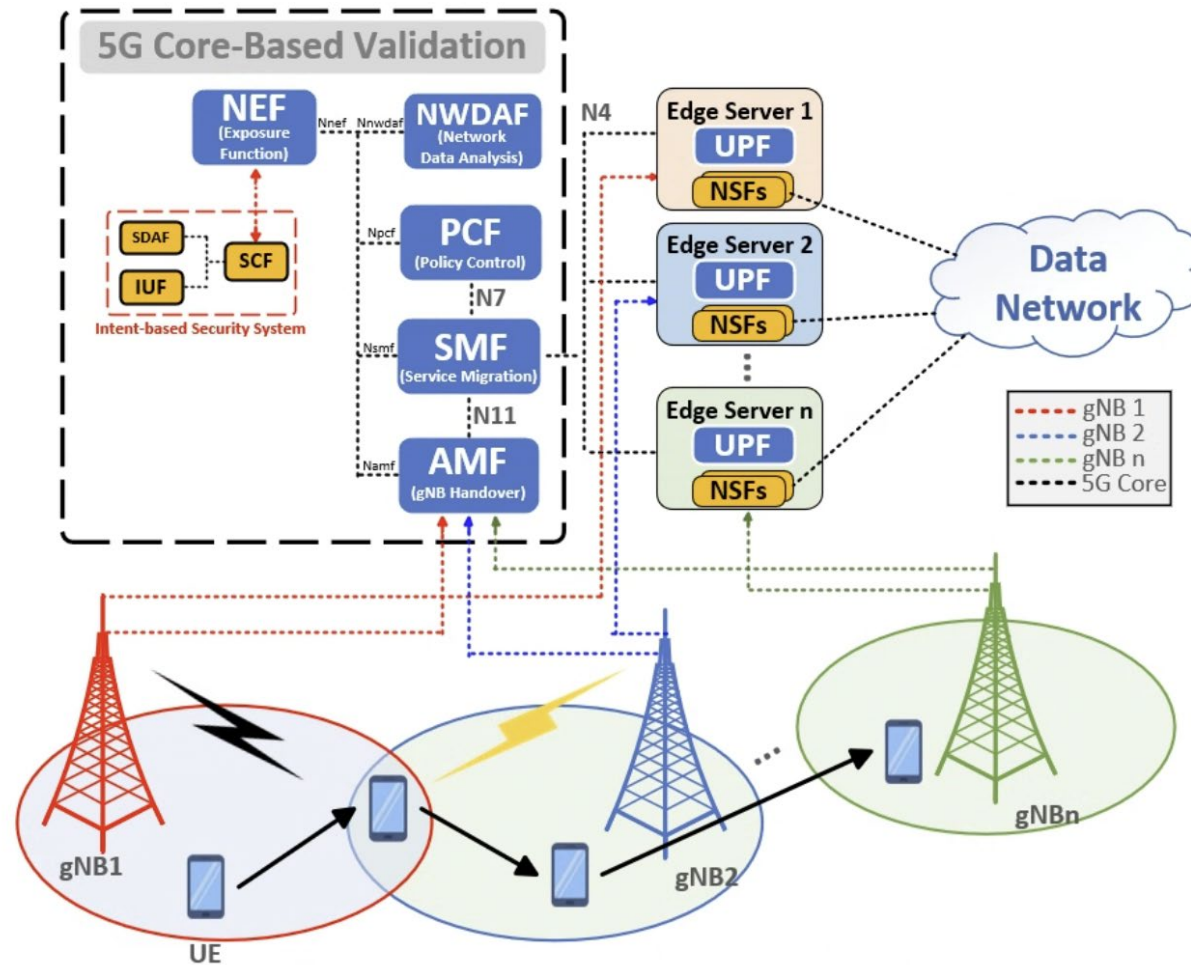
- **Legacy:** Execution of NSFs in Cloud server outside the 5G Core Networks
 - A detoured path from the UE to the cloud server causes **long delays** for security services.





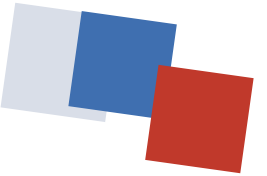
Edge-based Security Approach for 5G Networks

- **Proposed:** Execution of NSFs in Edge server in the 5G Core Networks
 - Short delay by an optimal path from a UE to Edge server for security services.





Open-Source Project for 5G-I2NSF



[URL] <https://github.com/jaehoonpauljeong/5G-I2NSF/tree/main/IETF-126>

 **5G-I2NSF** Public Watch 0

main 1 Branch 0 Tags Add file <> Code

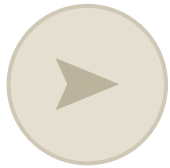
 **ahnjs124** Revise README to reflect project updates and goals 65b9fcd · now 28 Commits

 IETF-124	Add files via upload	9 months ago
 IETF-125	Add files via upload	4 months ago
 IETF-126	Add files via upload	2 days ago
 5G-I2NSF System Architecture.png	Add files via upload	6 minutes ago
 5G-I2NSF System.png	Add files via upload	9 months ago
 README.md	Revise README to reflect project updates and goals	now

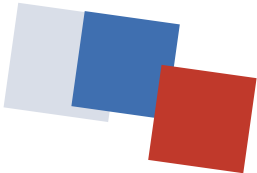
README  

5G-I2NSF

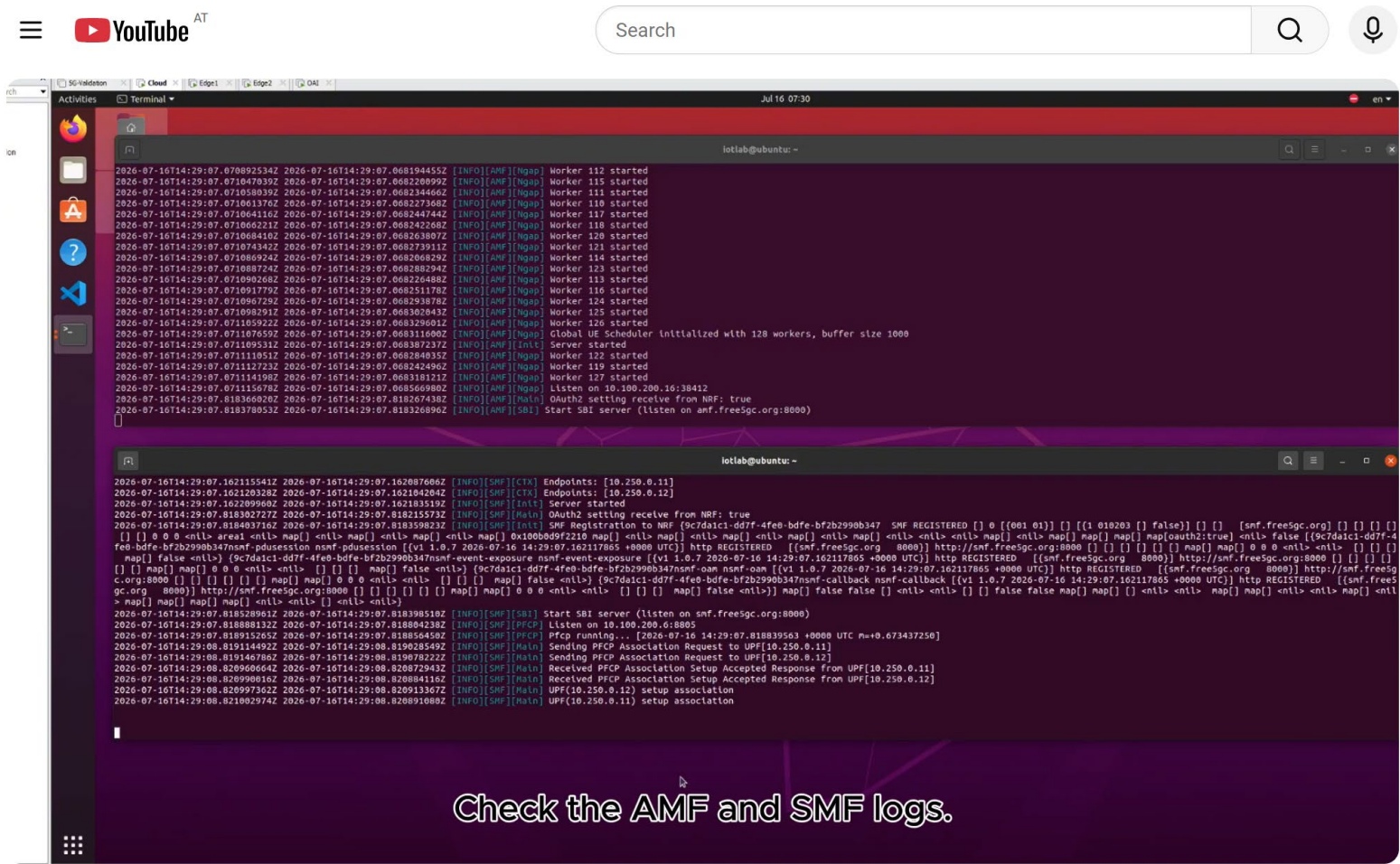
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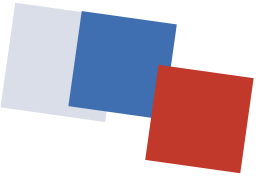
Demonstration Video Clip for 5G-I2NSF



[URL] <https://www.youtube.com/watch?v=uzc5LvEuX6c>



Check the AMF and SMF logs.



Cloud and Edge Security Approaches

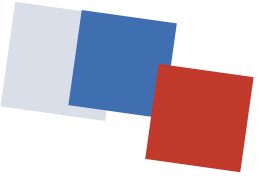
- This slide compares cloud-based and edge-based security approaches in terms of latency, mobility support, and deployment suitability.

Item	Cloud Approach	Edge Approach
NSF Location	Cloud server outside the 5G Core	Edge server near UPFs
Traffic Path	Detoured path to the cloud	Optimized path to the edge
Latency	Relatively high	Relatively low
Mobility Support	Difficult to maintain policy continuity during handover	Supports policy migration with UPF switching

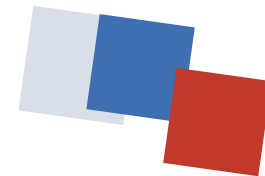
The edge approach is more suitable for latency-sensitive and mobility-aware 5G security services.



Conclusion and Summary



- The proposed 5G-I2NSF framework coordinates I2NSF security policy management with 5G mobility and user-plane control.
- During N2-based handover, the NSF policy is migrated from the Source Edge Server to the Target Edge Server through the SCF, helping maintain security continuity during UE mobility.
- The SDAF collects enforcement reports from the Target NSF for policy validation and security analysis.



Thank you